higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE

MATHEMATICS N5

(16030175)

31 July 2025 (X-paper)
09:00–12:00

Scientific calculators and drawing instruments may be used.

This question paper consists of 6 pages and a formula sheet of 5 pages.

tvetpro.co.za

066Q1G3125

Copyright reserved

GET IT ON Google Play

Try our Apps on Play Store

NATED Question Papers	**N1-N6** Mathematics NATED	**N1-N4** Engineering Science NATED	**N1-N6** Industrial Electronics NATED	**N4-N6** Electrotechnics NATED	**N4-N6** Mechanotechnics NATED
N5-N6 Power Machines NATED	**N4-N6** Quantity Surveying NATED	**N5-N6** Strength of Materials & Structures NATED	**N1-N3** Engineering Drawing NATED	**N3** Eletro-technology NATED	**N3** Mechano-technology NATED
N4-N6 Building Administration NATED	**N4-N6** Building& Structural Surveying NATED	**N4-N6** Building & Structural Construction NATED	**N1-N3** Building Science NATED	**N1-N3** Building Drawing NATED	**N4** Mechanical Draughting NATED
N1-N3 Electrical Trade Theory NATED	**N2-N3** Diesel Trade Theory NATED	**N5-N6** Strength of Materials & Structures NATED	**N3** Building & Civil Technology NATED	**N1-N2** Fitting and Machining Theory NATED	**N1-N4** Plating and Structural Steel Drawing NATED
N1-N3 Motor Trade Theory NATED	**N5-N6** Mechanical Drawing & Design NATED	**N1-N2** Plumbing Theory NATED	**N4-N6** Fault Finding & Protective Devices NATED	**N2-N3** Instrument Trade Theory NATED	**N5-N6** Engineering Physics NATED
N5-N6 Fluid Mechanics NATED	**N4-N6** Supervisory Management NATED	**N4-N6** Digital Electronics NATED	**N6** Control Systems NATED	**N1-N2** Bricklaying &Plastering Theory NATED	More…

DEPARTMENT OF HIGHER EDUCATION AND TRAINING
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE
MATHEMATICS N5
TIME: 3 HOURS
MARKS: 100

INSTRUCTIONS AND INFORMATION

1. Answer all the questions.
2. Read all the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. Show all intermediate steps and simplify where possible.
5. All final answers must be rounded off to THREE decimals.
6. Questions may be answered in any order, but subsections of questions must be kept together.
7. Sketches must be large, neat and fully labelled.
8. Use only a black or blue pen.
9. Write neatly and legibly.

---

tvetpro.co.za

Copyright reserved

QUESTION 1

1.1 Determine the following limit:

$$y=\lim_{x\to\infty}\frac{x}{\ln^3(2x-1)} \tag{4}$$

1.2 Given:

$$\cos x=\lim_{x\to 0}\frac{\left(\tan^{-1}x\right)^2}{x}$$

Calculate the numerical value of $\cos x$. (2)

[6]

QUESTION 2

2.1 Given:

$$f(x)=\frac{8}{\sqrt{x}}$$

Determine the simplest form of:

2.1.1 the first three terms of $f(x+h)$ (2)

2.1.2 $f(x+h)-f(x)$ (1)

2.1.3 $\dfrac{f(x+h)-f(x)}{h}$ (1)

2.1.4 $\lim\limits_{h\to 0}\dfrac{f(x+h)-f(x)}{h}$ (1)

2.2 Determine $\dfrac{dy}{dx}$ in each of the following cases (simplification not required):

2.2.1 $y=(5+\log_4 x)^4(7x-e^{\sin 4x})^5$ (4)

2.2.2 $y=\tan^3\left[\sin\left(\dfrac{7x^2+2}{x-3}\right)\right]$ (5)

2.3 Calculate $\dfrac{dy}{dx}$ if $y=(5-4x)^{\frac{1}{x}}$ with aid of logarithmic differentiation. (4)

tvetpro.co.za

2.4 Given the implicit function:

$$x^3+5xy-2y^2=8$$

2.4.1 Determine $\frac{dy}{dx}$ (3)

2.4.2 Determine the equation of the tangent to the graph at the point $(1;2)$ (3)

[24]

QUESTION 3

3.1 Given: $f(x)=x^3-2x^2+4$

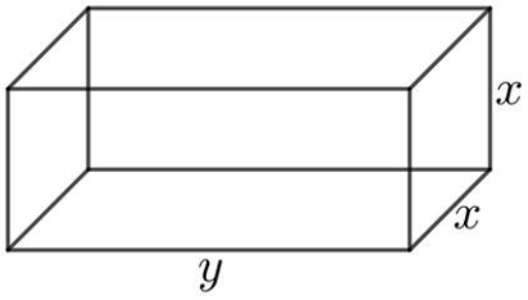
3.1.1 Determine the coordinates of the turning points of $f(x)$. (3)

3.1.2 Draw up a table of x and $f(x)$, where x is ranging from $x=-2$ to $x=3$, $x\epsilon\ Z$. (2)

3.1.3 Draw a neat graph of $f(x)$ between these values, show the turning points on it. (2)

3.1.4 If $x=-1{,}2$ is an estimated value of the root of the equation $x^3-2x^2+4=0$, use Taylor's/Newton's method TWICE to determine a better approximation of this root. (Correct root to THREE decimal figures.) (4)

3.2 A rectangular shell has edges welded from steel bars, with square ends. If the total length of the steel bars is 144 metres, find the dimensions of the shell that yields the maximum volume.



(5)

3.3 The surface area of a snowball decreases at a rate of $6\,m^2/hr$.

How fast is the diameter changing when the radius is $2\ m$? (4)

HINT: $A=4\pi r^2$

[20]

tvetpro.co.za

QUESTION 4

4.1 Determine $\int y \ dx$ in each of the following cases:

4.1.1 $y=\dfrac{2x-1}{(2x^2-2x+5)^2}$ (3)

4.1.2 $y=\sin^5 x\cos^3 x$ (5)

4.1.3 $y=\dfrac{2x^4+x^3+x^2-2}{2x+3}$ (6)

4.1.4 $y=x\log_3 4x$ (4)

4.2 Determine $\int y \ dx$ by resolving the integral into partial fractions:

$$y=\frac{2x+7}{x^2+x-2}$$ (5)

[23]

QUESTION 5

5.1 The time (t) and velocity (v) of an object moving in a medium at time t is given by:

$$t=\int_{10}^{100}\frac{1}{v(2v+1)}dv$$

Evaluate t. (5)

5.2 Given the curves: $y=x^2$ and $y=2-x$

5.2.1 Calculate the coordinates of the points of intersection of $y=x^2$ and $y=2-x$.

5.2.2 Make a neat sketch to show the enclosed area, the representative strip, and the points of intersection.

5.2.3 Calculate the magnitude of the area in QUESTION 5.2.2.

(3 × 3) (9)

5.3 Calculate the moment of inertia of a uniform circular wheel of diameter 300 cm about an axis through its centre and perpendicular to the wheel. The mass of the wheel is 50 kg. (5)

[19]

tvetpro.co.za

QUESTION 6

6.1 Determine the general solution of:

$$\frac{dy}{dx}=\frac{(x-1)y^5}{x^2(2y^3-y)} \qquad (3)$$

6.2 Determine the particular solution of:

$$\frac{d^2y}{dx^2}-9x-12\sin 3x=0, \quad \text{with} \quad \frac{dy}{dx}=2 \text{ and } y=1 \text{ at } x=0 \qquad (5)$$

[8]

TOTAL: 100

tvetpro.co.za

Copyright reserved

FORMULA SHEET

Any applicable formula may also be used.

TRIGONOMETRY

$\sin^2 x + \cos^2 x = 1$

$1 + \tan^2 x = sec^2 x$

$1 + \cot^2 x = cosec^2 x$

$\sin 2A = 2\sin A \cos A$

$\cos 2A = \cos^2 A - \sin^2 A$

$$\tan 2A = \frac{2\tan A}{1 - \tan^2 A}$$

$$\sin^2 A = \frac{1}{2} - \frac{1}{2}\cos 2A$$

$$\cos^2 A = \frac{1}{2} + \frac{1}{2}\cos 2A$$

$$\sin(A \pm B) = \sin A \cos B \pm \sin B \cos A$$

$$\cos(A \pm B) = \cos A \cos B \pm \sin A \sin B$$

$$\tan(A \pm B) = \frac{\tan A \pm \tan B}{1 \pm \tan A \tan B}$$

$$\sin A \cos B = \frac{1}{2}[\sin(A + B) + \sin(A - B)]$$

$$\cos A \sin B = \frac{1}{2}[\sin(A + B) - \sin(A - B)]$$

$$\cos A \cos B = \frac{1}{2}[\cos(A + B) + \cos(A - B)]$$

$$\sin A \sin B = \frac{1}{2}[\cos(A - B) - \cos(A + B)]$$

$$\tan x = \frac{\sin x}{\cos x}; \sin x = \frac{1}{cosec\ x}; \cos x = \frac{1}{sec\ x}$$

tvetpro.co.za

tvetpro.co.za

BINOMIAL THEOREM

$$(x+h)^n = x^n + n\,x^{n-1}\;h + \frac{n(n-1)}{2!}x^{n-2}\;h^2 + \ldots$$

DIFFERENTIATION

$$e = -\frac{f(a)}{f'(a)}$$

$$r = a + e$$

PRODUCT RULE

$$y = u(x)\;.\;v(x)$$

$$\frac{dy}{dx} = u\;.\;\frac{dv}{dx} + v\;.\;\frac{du}{dx}$$

$$= u\;.\;v' + v\;.\;u'$$

QUOTIENT RULE

$$y = \frac{u(x)}{v(x)}$$

$$\frac{dy}{dx} = \frac{v\;.\;\frac{du}{dx} - u\;.\;\frac{dv}{dx}}{v^2}$$

$$\frac{v\;.\;u' - u\;.\;v'}{v^2}$$

CHAIN RULE

$$y = f(u(x))$$

$$\frac{dy}{dx} = \frac{dy}{du}\;.\;\frac{du}{dx}$$

tvetpro.co.za

Copyright reserved

$f(x)$	$\frac{d}{dx}f(x)$	$\int f(x)dx$
ax^n	nax^{n-1}	$\frac{ax^{n+1}}{n+1}+c$
a	0	$ax+c$
e^x	e^x	e^x+c
a^x	$a^x.\ lna$	$\frac{a^x}{lna}+c$
$\log_e x$	$\frac{1}{x}$	—
$\log_a x$	$\frac{1}{xlna}$	—
$\sin x$	$\cos x$	$-\cos x+c$
$\cos x$	$-\sin x$	$\sin x+c$
$\tan x$	$sec^2 x$	$\ln(sec\,x)+c$
$\cot x$	$-cosec^2 x$	$\ln(\sin x)+c$
$sec\,x$	$sec\,x\ .\tan x$	$\ln[sec\,x+\tan x]+c$
$cosec\,x$	$-cosec\,x\ .\ \cot x$	$\ln(cosec\,x-\cot x)+c$
$\sin^{-1}x$	$\frac{1}{\sqrt{1-x^2}}$	—
$\cos^{-1}x$	$\frac{-1}{\sqrt{1-x^2}}$	—
$\tan^{-1}x$	$\frac{1}{1+x^2}$	—
$\cot^{-1}x$	$\frac{-1}{1+x^2}$	—
$sec^{-1}x$	$\frac{1}{\sqrt{x^2-1}}$	—
$cosec^{-1}x$	$\frac{-1}{x\sqrt{x^2-1}}$	—

tvetpro.co.za

Copyright reserved

tvetpro.co.za

$f(x)$	$\frac{d}{dx}f(x)$	$\int f(x)dx$
$\frac{1}{\sqrt{a^2-x^2}}$	—	$\sin^{-1}\left(\frac{x}{a}\right)+c$
$\frac{1}{a^2+x^2}$	—	$\frac{1}{a}\tan^{-1}\left(\frac{x}{a}\right)+c$
$\frac{1}{x\sqrt{x^2+a^2}}$	—	$\frac{1}{a}sec^{-1}\left(\frac{x}{a}\right)+c$
$\sqrt{a^2-x^2}$	—	$\frac{a^2}{2}\sin^{-1}\left(\frac{x}{a}\right)+\frac{x}{2}\sqrt{a^2-x^2}+c$
$\frac{1}{x^2-a^2}$	—	$\frac{1}{2a}\ln\left(\frac{x-a}{x+a}\right)+c$
$\frac{1}{a^2-x^2}$	—	$\frac{1}{2a}\ln\left(\frac{a+x}{a-x}\right)+c$

INTEGRATION

$$\int f(x)\ g'(x)\ dx = f(x)\ g(x) - \int f'(x) g\ (x)dx$$

$$\int \frac{f'(x)}{f(x)}dx = \ln\ f(x)\ + c$$

$$\int [f(x)]^n\ f'(x)dx = \frac{[f(x)]^{n+1}}{n\ +1} + c$$

$$\frac{f(x)}{(ax+b)(cx\ +d)} = \frac{A}{ax\ +b} + \frac{B}{cx\ +d}$$

$$\frac{f(x)}{(x\ +a\ )^n} = \frac{A}{(x\ +a)} + \frac{B}{(x\ +a)^2} + \frac{C}{(x+a)^3} + \ldots \frac{Z}{(x\ +a)^n}$$

APPLICATIONS OF INTEGRATION

AREAS

$$A_x = \int_a^b ydx\ ;\ A_x = \int_a^b (y_1 - y_1)dx$$

tvetpro.co.za

$$A_x = \int_a^b x\,dy \;\; ; A_{Y=} \int_a^b (x_1 - x_2)\,dy$$

tvetpro.co.za

VOLUMES

$$V_x = \pi \int_a^b y^2 \, dx \; ; \; V_{X\,=\,\pi} \int_a^b \left(y_1^2 - y_2^2 \right) dx$$

$$V_y = \pi \int_a^b x^2 \, dy \; ; \; V_{y=\,\pi} \int_a^b \left(x_1^2 - x_2^2 \right) dy$$

SECOND MOMENT OF AREA

$$I_x = \int_a^b r^2 \, dA \; ; \; I_{y=} \int_a^b r^2 \, dA$$

MOMENT OF INERTIA

$Mass = density \times volume$

$M = pv$

$DEFINITION : I = m \; r^2$

$$GENERAL : I = \int_a^b r^2 dm = p \int_a^b r^2 dV$$

tvetpro.co.za

Copyright reserved